Embedded System HW Design
Week8
System Design Techniques

- Summarize how proper design methodology can speed up time to market and improve quality & cost
- Identify various Design Methodologies and understand which suits best for your needs
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Week8
Design Methodologies

- Process for creating a system
- **Many Embedded Systems are complex**
  - Large specifications
  - Multiple Designers
  - Interface to Manufacturing

- **Proper processes improve**:
  - Quality
  - Cost of Design and Manufacturing

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Product Metrics

- **Time to Market**
  - Beat competitors to market
  - Meet market window(back to school)
- Design Cost
- Manufacturing Cost
- Quality

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Spacecraft design errors

- **Mars Climate Observer:**
  Lost on Mars in September 1999.
- **Requirements problem**:
  - Requirements did not specify units.
  - Lockheed Martin used English; JPL wanted metric.
- Not caught by manual inspections.

- **New Horizons:**
  Experienced 81-minute comm blackout while approaching Jupiter in 2015.
- **LightSail:**
  Malfunctioned in orbit due to a file overflow bug.
  - Bug occurred only after about 40 hours of operation; ground tests did not run long enough.

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Design Flow

- **Design Flow**: Sequence of steps in a design methodology
- May be partially or fully automated
  - Use tools to transform, verify design
- Design flow is one component of methodology. Methodology also includes management organization, etc

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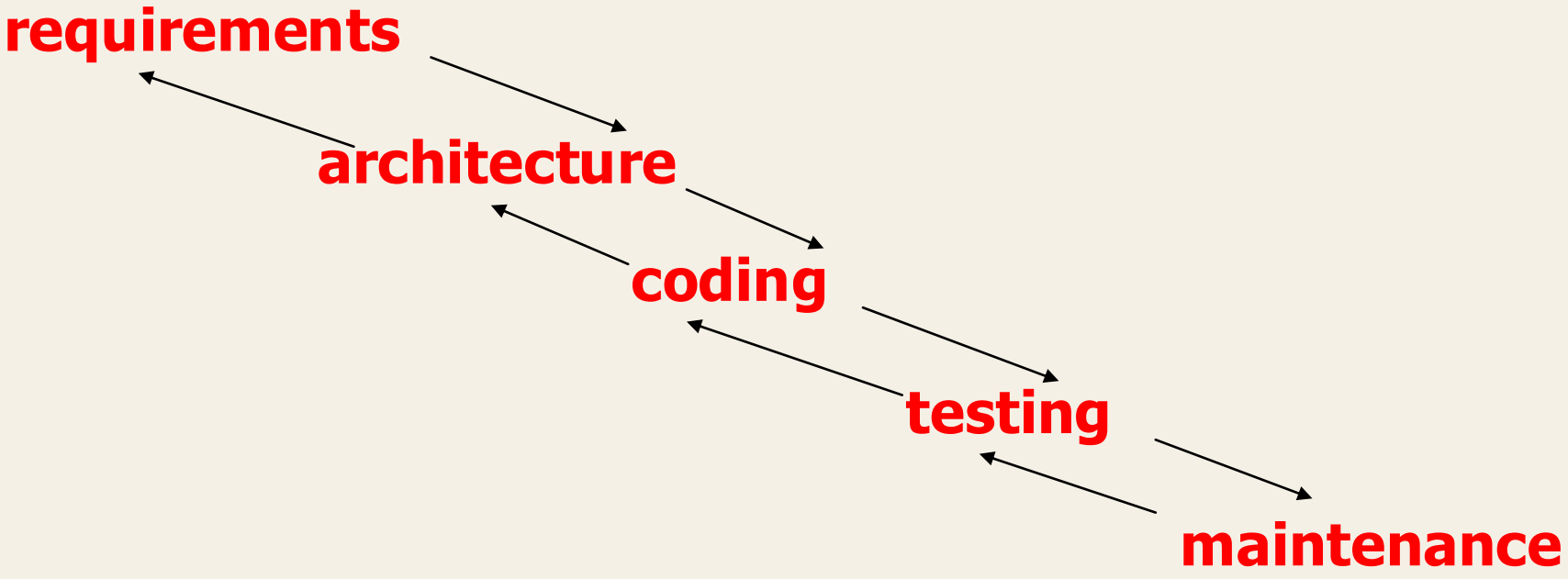
Waterfall Model

- Early model for software development



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Waterfall Model steps

- **Requirements**: Determine basic characteristics
- **Architecture**: Decompose into basic modules
- **Coding**: Implement and Integrate
- **Testing**: Exercise and uncover bugs
- **Maintenance**: Deploy, fix bugs, upgrade

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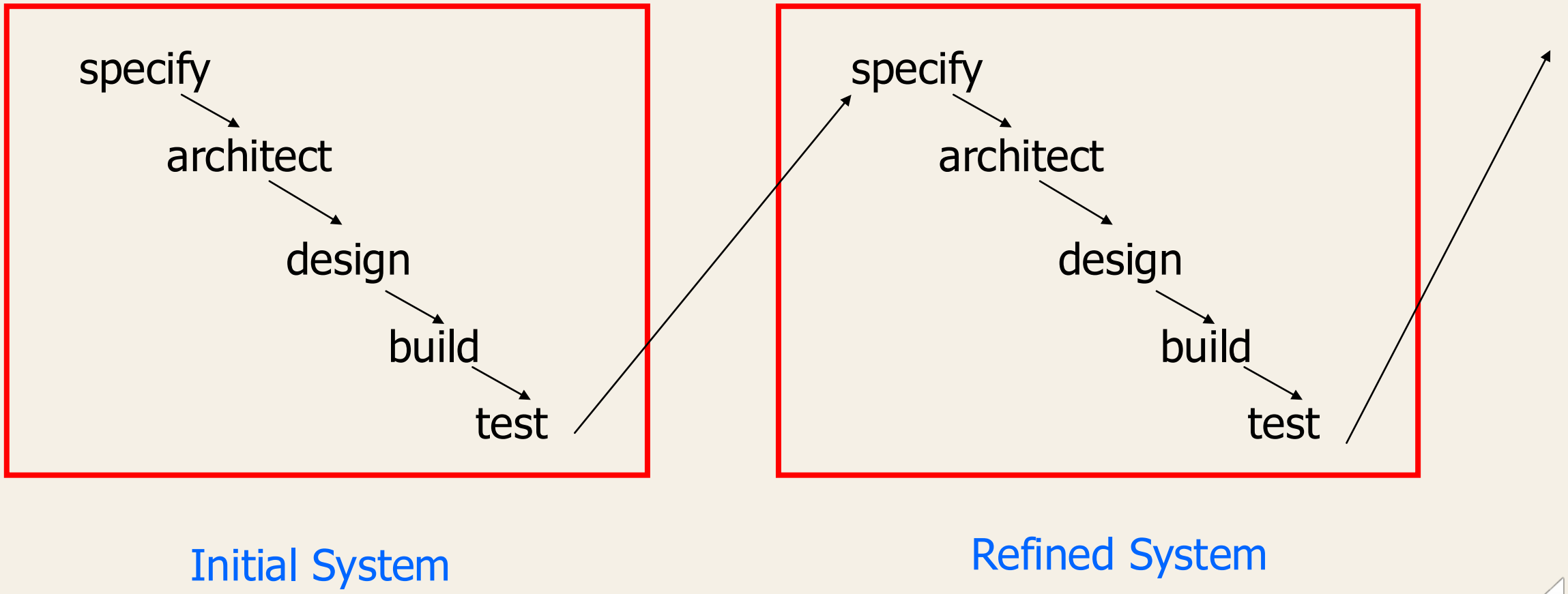
Successive Refinement Model



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Agile software design

- Minimize time, cost of deployment.
  - Reduced or eliminated requirements process.
  - Reduced documentation.
  - Simplified testing.
- Often used in consumer-oriented software.
- Does not provide documentation, testing required in many real-time embedded system applications.

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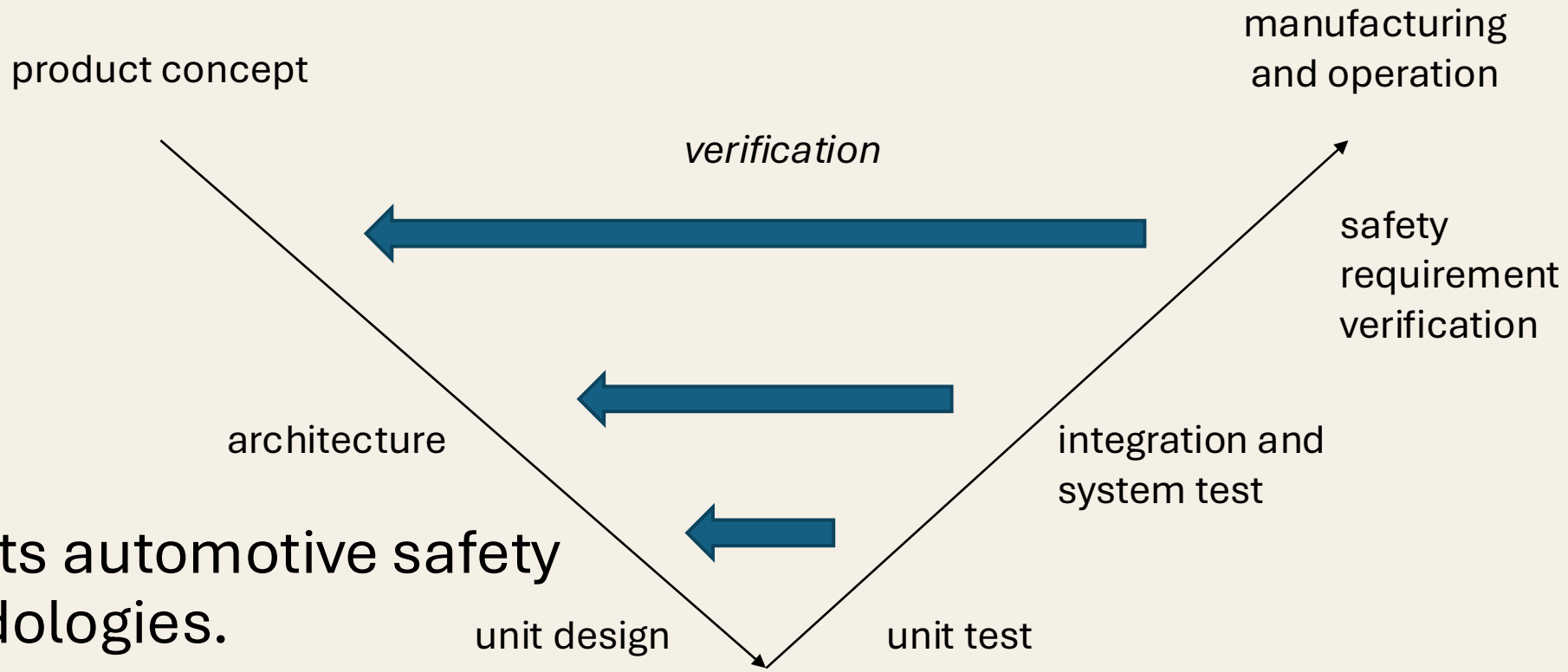
V model



- Supports automotive safety methodologies.
- Top-down design followed by bottom-up verification.

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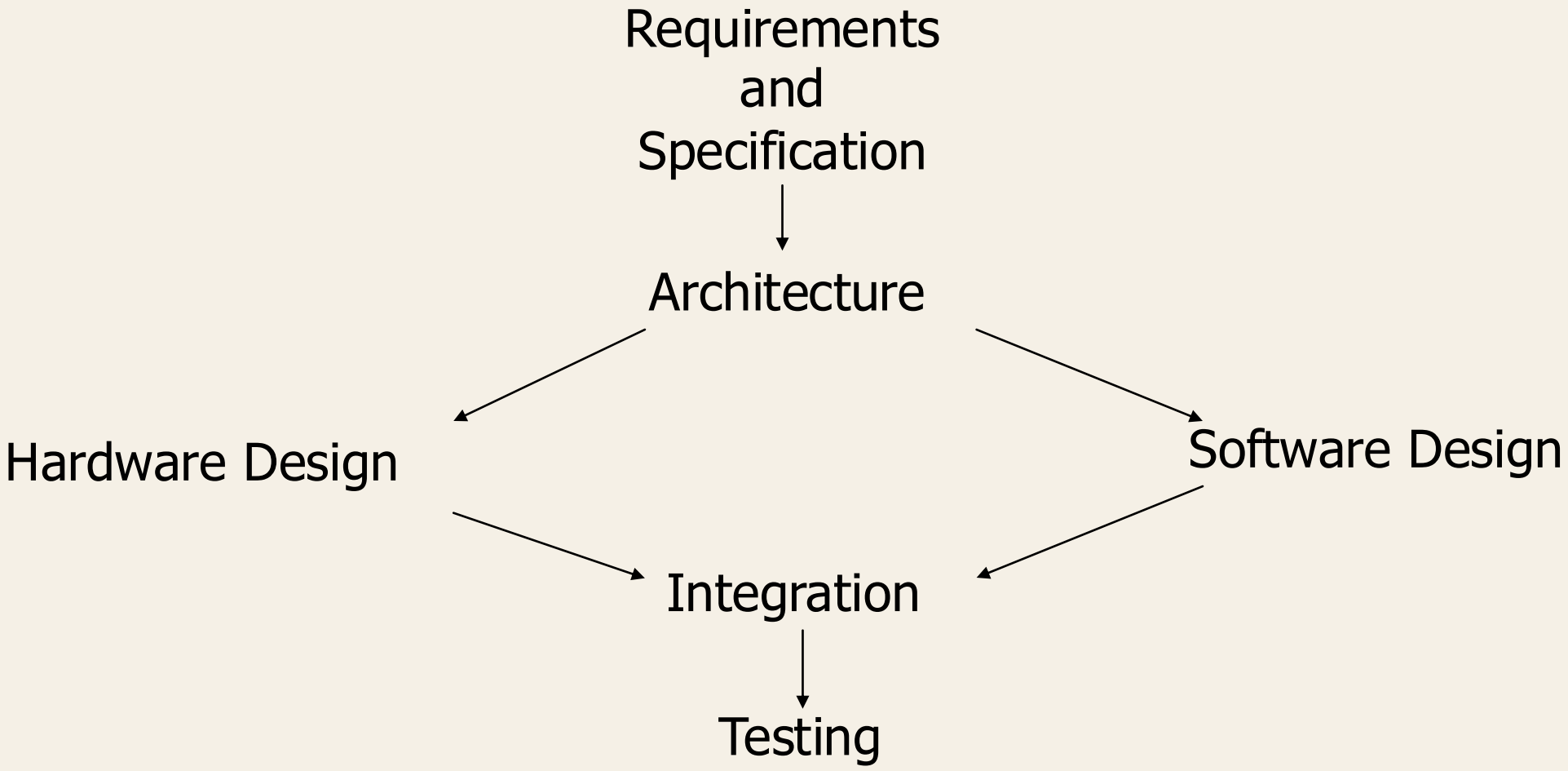
Hardware/Software Design Flow



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Hierarchical Design Flow

- **Embedded systems must be designed across multiple levels of abstraction**:
  - System Architecture
  - Hardware and Software systems
  - Hardware and Software components

- Often need design flows within design flows

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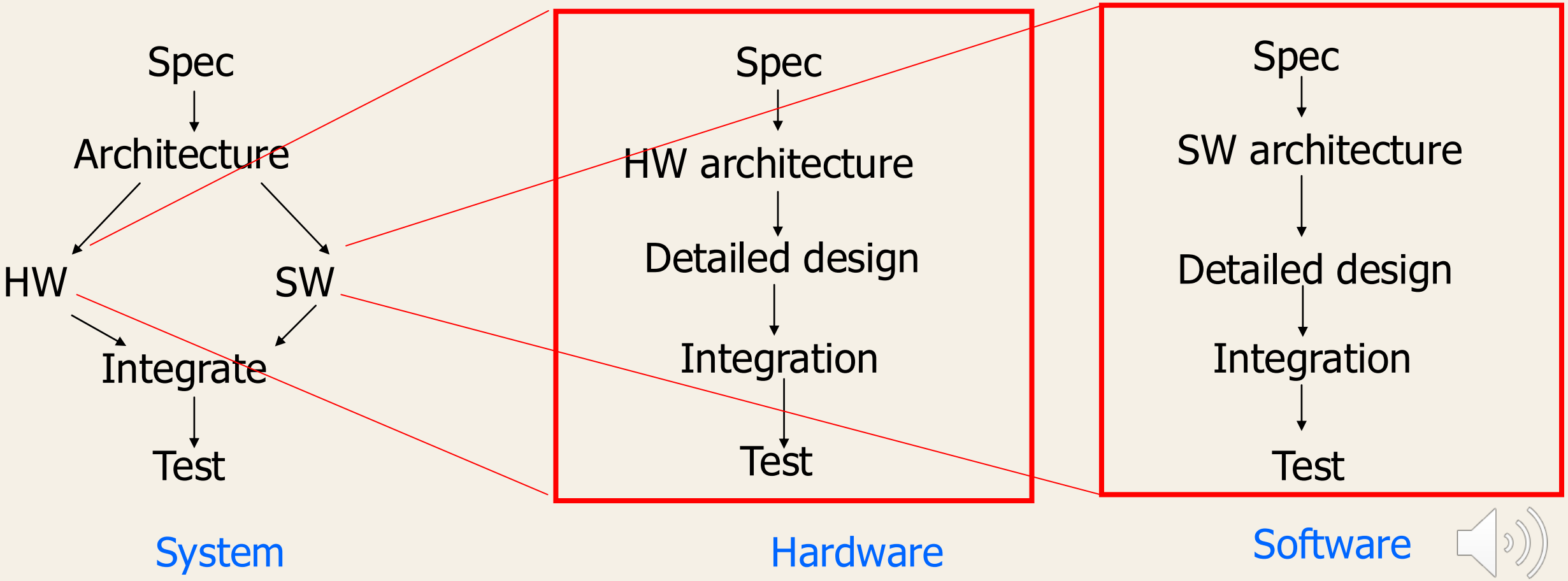
Hierarchical HW/SW Flow



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Concurrent Engineering

- Large projects use many people from multiple disciplines
- Work on several tasks at once to reduce design time
- Feedback between tasks helps improve quality, reduce number of later design problems

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Requirements Analysis

- Requirements: Informal description of what customer wants
- Specification: Precise description of what design team should deliver
- Requirements phase links customers with designers

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Types of Requirements

- Functional: Input/Output relationships

- Non-functional:
  - Timing
  - Power consumption
  - Manufacturing cost
  - Physical size
  - Time-to-market
  - Reliability

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Good Requirements

- Correct
- Unambiguous
- Complete
- Verifiable: Is each requirement satisfied in the final system?
- Consistent: Requirements do not contradict each other

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Good Requirements(Contd)

- **Modifiable**: Can update requirements easily

- **Traceable**:
  - Know why each requirement exists
  - Go from source documents to requirements
  - Go from requirement to implementation
  - Back from implementation to requirement

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Settling Requirements

- Customer interviews
- Comparison with competitors
- Sales feedback
- Mock-ups, prototypes

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Quality Assurance

ISO 9000

- Developed by International Standards organization
- Applies to a broad range of industries
- Concentrates on the process
- Validation based on extensive documentation of organization's process

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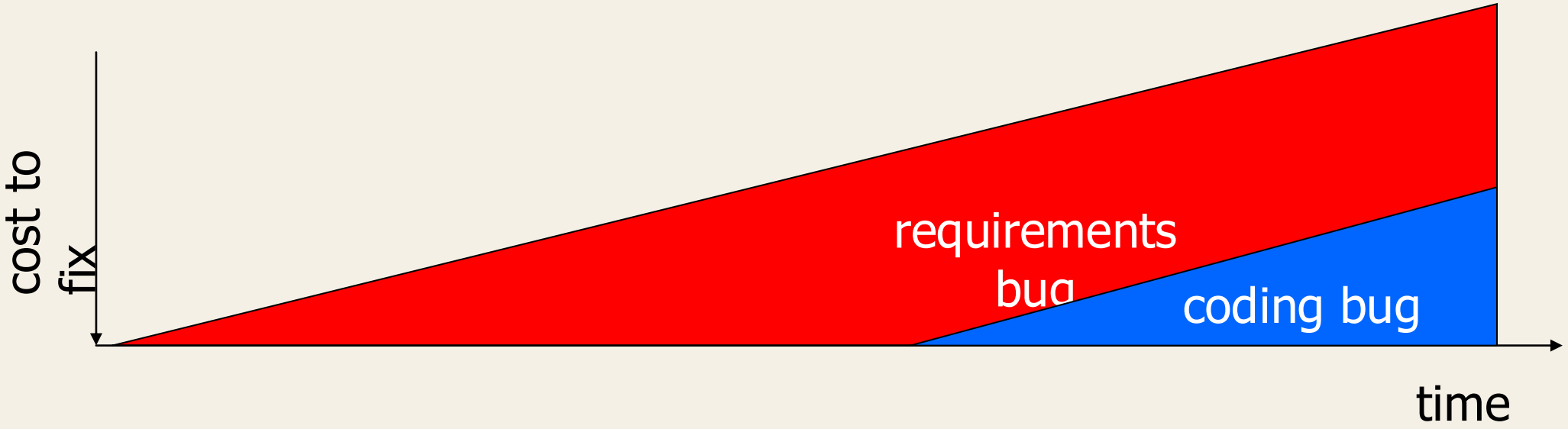
Verification

Verification and testing are important throughout the design flow.

Early bugs are more expensive to fix:



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Verification Requirements and Specification

- **Requirements**:
  - Prototypes
  - Prototyping languages
  - Pre-existing systems
- **Specifications**:
  - Usage scenarios
  - Formal techniques

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Design Review

- Uses meetings to catch design flaws
  - Simple, low-cost
  - Proven by experiments to be effective
- Use other people in the project/company to help spot design problems

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Design Review Players

- **Designers**: Present the design to the rest of the team, make changes
- **Review Leader**: Coordinates process
- **Review Scribe**: Takes notes of meetings
- **Review Audience**: Looks for bugs

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Follow Up

- Designers make suggested changes.
  - Document changes

- Leader checks on results of changes,
  may distribute to audience for further review or additional reviews

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